ITEC634 – Week 6 Lab 3 Report

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Week 6 Lab 3 – ASP.NET Core Razor Pages Multi-Page Application

1. Introduction

This lab focused on developing a multi-page web application using ASP.NET Core Razor Pages and Bootstrap. The objective was to design a structured web application with multiple pages, implement clean navigation using a shared layout, and demonstrate dynamic routing using URL parameters. The lab provided practical exposure to building scalable web applications with consistent user interface design and responsive behavior across devices.

2. Objectives of the Lab

The main objectives of this lab were:

- To understand how to build a multi-page ASP.NET Core Razor Pages application
- To implement a shared layout with a consistent navigation bar across all pages
- To create and manage multiple pages (Home, Product Catalog, About)
- To implement dynamic routing using URL parameters
- To apply Bootstrap for responsive and modern UI design

3. Implementation Process

Project Setup

The application was developed using Visual Studio Code and the .NET SDK. A new Razor Pages project was created using the `dotnet new webapp` command. This automatically generated the project structure, including the Pages directory and shared layout files.

Layout and Navigation Design

The `_Layout.cshtml` file was modified to implement a site-wide navigation bar using Bootstrap. The navbar includes links to Home, Product Catalog, and About pages, ensuring a consistent user experience across all pages.

Home Page (Index.cshtml)

The Home page was designed as a welcoming interface using a Bootstrap hero section. It provides an introduction to the application and includes a button that directs users to the Product Catalog page.

Product Catalog Page (Products.cshtml)

A new page was created to display a list of products in a structured table format. The table includes product names, prices, and a "View" link for each item. Bootstrap classes were used to enhance the table's appearance and responsiveness.

Dynamic Routing (Details Page)

A dynamic details page was implemented using route parameters (`@page "{id:int}"`). When a user clicks on a product, they are redirected to a URL such as `/Details/1`. The product ID is passed to the backend and displayed dynamically on the page, demonstrating how data can be transferred through URLs.

About Page (About.cshtml)

An About page was created to provide a brief description of the application. This page completes the multi-page requirement and demonstrates proper routing within the application.

Displaying Results

The result was displayed dynamically on the same page using Razor syntax. The message was shown below the form after submission, ensuring that the application met the requirement of a single-page interaction.

4. Key Features of the Application

- Multi-page navigation using Razor Pages
- Shared layout with a consistent Bootstrap navigation bar
- Product catalog displayed in a responsive table
- Dynamic routing using URL parameters
- Clean and user-friendly interface
- Fully responsive design across different screen sizes

5. Challenges Encountered and Solutions

During the development process, several challenges were encountered:

- Navbar Not Updating

Initially, changes made to the `_Layout.cshtml` file did not reflect in the browser. This was resolved by ensuring the file was properly saved and restarting the application using `dotnet run`.

- Application Lock Issue

At one point, the terminal became unresponsive because the application was already running. This was resolved by stopping the running process using `CTRL + C`.

- Routing Errors

Errors occurred when accessing the Details page due to incorrect route configuration. This was fixed by correctly defining the route parameter using `@page "{id:int}"` and ensuring the backend method accepted the parameter.

6. What I Learned

This lab significantly improved my understanding of building structured web applications using ASP.NET Core Razor Pages. I learned how to design multi-page applications with shared layouts and implement navigation that enhances user experience.

I also gained practical knowledge of dynamic routing, which allows data to be passed through URLs and processed on different pages. Additionally, I developed a better understanding of responsive design using Bootstrap and how web applications adapt to different screen sizes and devices.

The lab also improved my debugging skills, particularly in resolving issues related to routing, layout updates, and application execution. Overall, this lab strengthened my ability to develop modern, scalable web applications.Overall, this lab provided a strong foundation for building interactive web applications using ASP.NET.

7. Use of Artificial Intelligence (AI)

AI tools such as ChatGPT were used to support the development of this lab. The AI assisted in:

- Understanding Razor Pages structure and routing
- Debugging layout and navigation issues
- Implementing dynamic routing and page connections
- Providing step-by-step guidance for building the application
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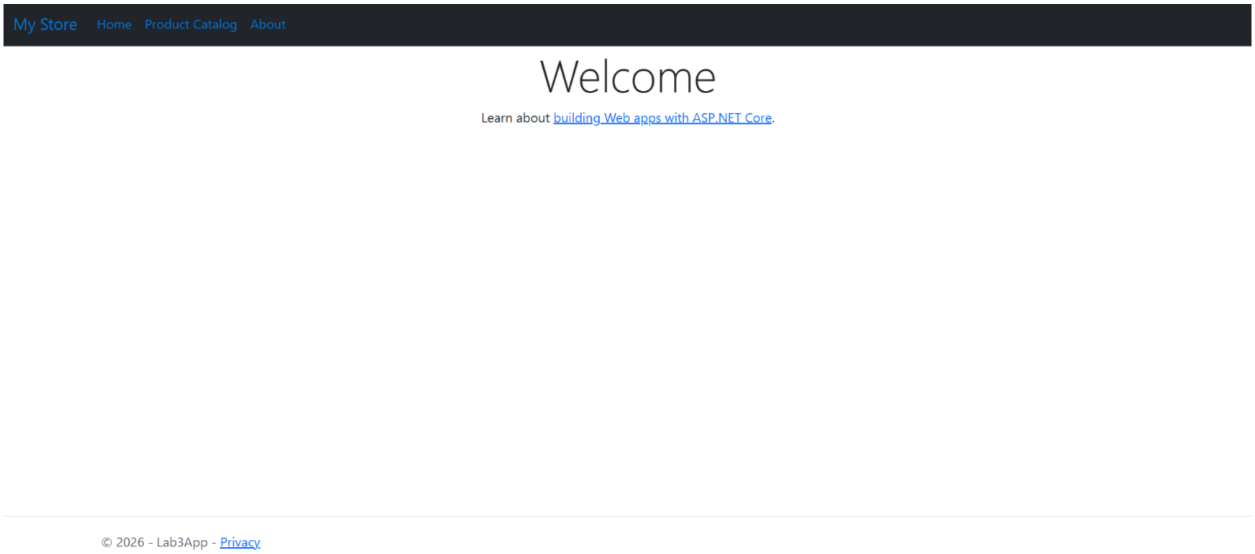
Summary of Prompts Used

- How to create a multi-page Razor Pages application
- How to implement a shared layout and navigation bar
- How to create dynamic routes using URL parameters
- Debugging issues related to routing and layout updates

Summary of AI Responses

The AI provided structured explanations, code examples, and troubleshooting guidance. It helped simplify complex concepts such as routing and layout management, making the implementation process easier and more efficient.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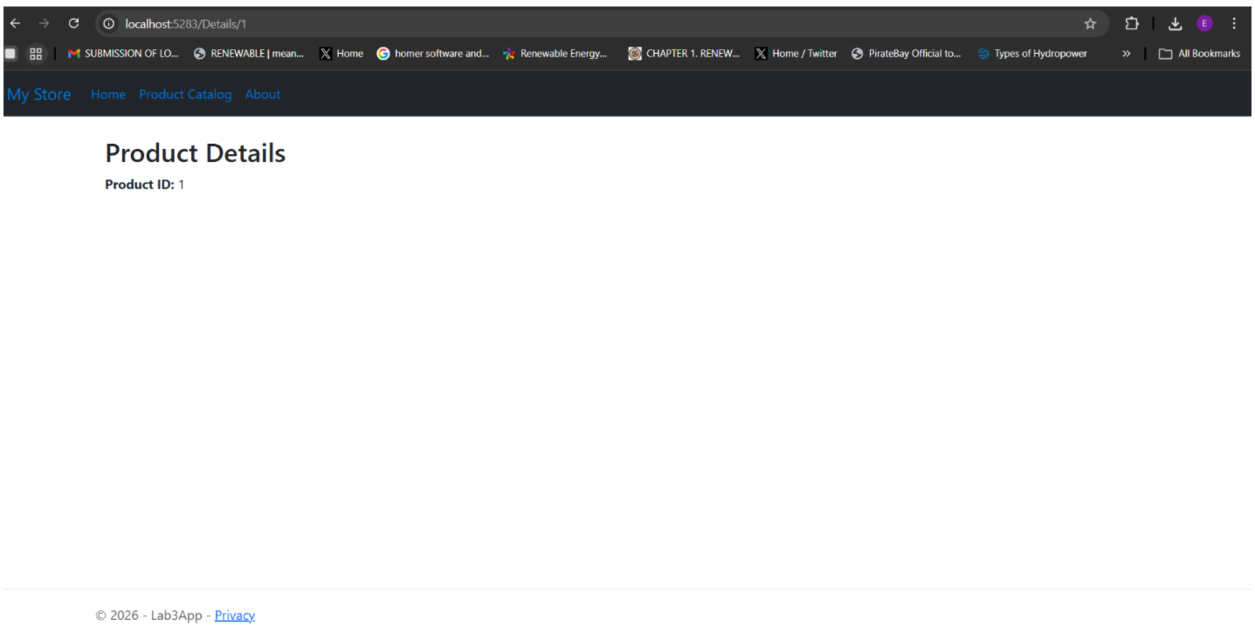
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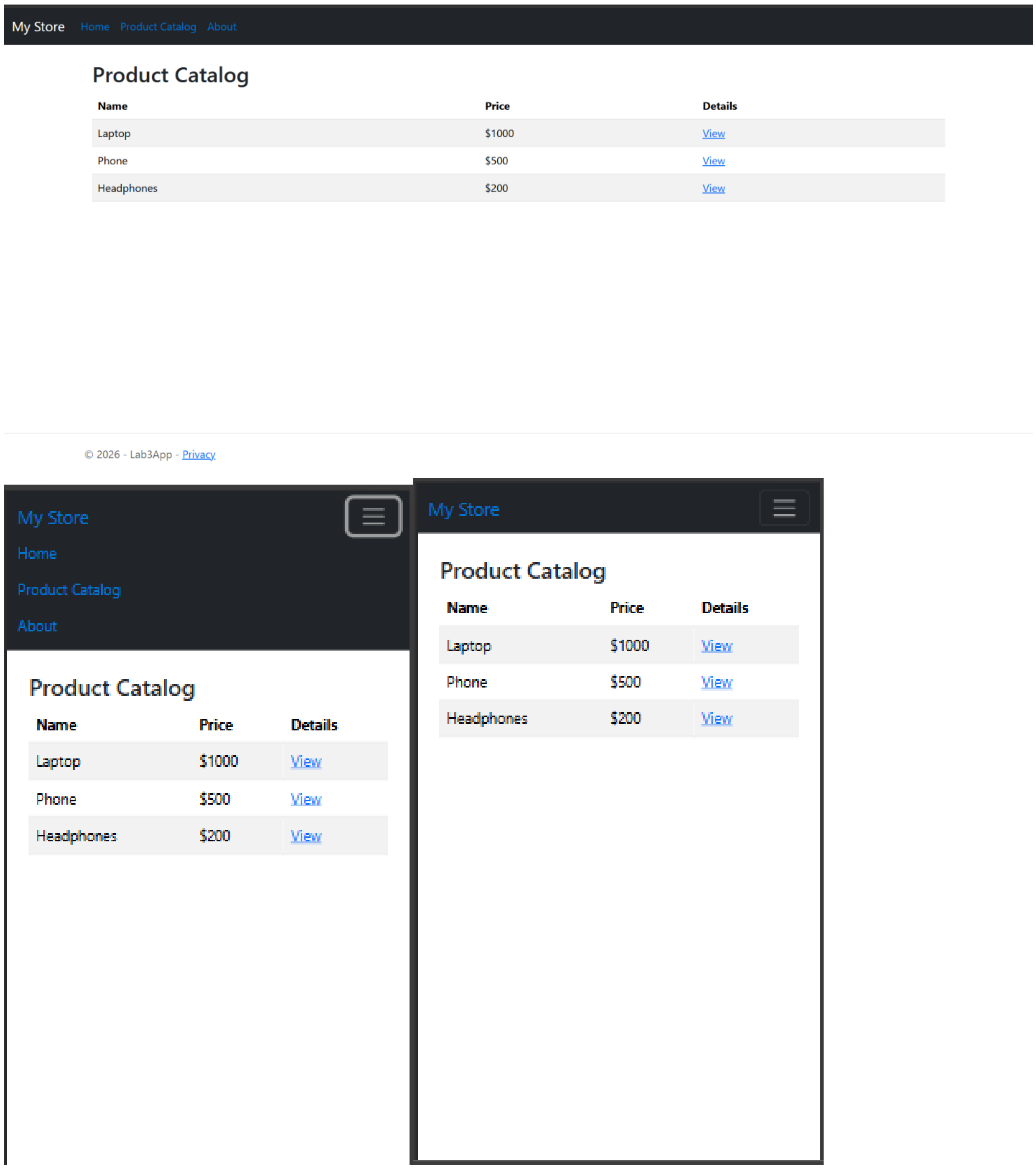

Screenshots of the Application in Web and Mobile Version

8. Conclusion

In conclusion, this lab successfully demonstrated how to build a multi-page web application using ASP.NET Core Razor Pages. The objectives of implementing shared navigation, creating multiple pages, and using dynamic routing were achieved.

The application also met the requirement for responsive design using Bootstrap, ensuring usability across different devices and browsers. This lab has enhanced my practical skills in web development and provided valuable experience in designing modern web applications. The knowledge gained will be beneficial for future projects involving web technologies and application development.